# Semisupervised Band Selection With Graph Optimization for Hyperspectral Image Classification

Fang He, Feiping Nie, *Member, IEEE*, Rong Wang, *Member, IEEE*, Weimin Jia, Fenggan Zhang, and Xuelong Li, *Fellow, IEEE*

***Abstract*— Semisupervised band selection (BS) technique plays an important role in processing hyperspectral images (HSIs) because of its superiority of using the limited labeled data and plentiful unlabeled data to select the discriminative and informative feature, which copes with the problem of high-dimensional and scare labeled samples of HSIs. Among semisupervised BS models, graph-based methods are superior to others in many situations and have received more and more attention. However, traditional graph-based models construct a similarity matrix and select valuable bands independently. The similarity matrix remains constant, which will damage the local manifold structure and lead to a suboptimal result. To solve this problem, we propose a semisupervised band selection with an optimal graph (BSOG) approach, which performs BS and local structure learning simultaneously. Instead of fixing the input similarity matrix, the similarity matrix is updated constantly to learn a better local structure. Besides, the learned similarity matrix is adaptive. Then, the optimal band subset can be selected by analyzing the obtained projection matrix $W$. An efficient and simple optimization algorithm is proposed to solve this model. Experiments on four real HSIs data validate the effectiveness of the proposed model.**

***Index Terms*— Band selection (BS), hyperspectral images (HSIs), optimal graph, semisupervised.**

## I. Introduction

With the development of spectral imaging techniques in recent years, hyperspectral images (HSIs) have been applied in various field, such as military aerial surveillance, environment monitoring, precision agriculture, and resource management, as they contain hundreds of narrow spectral bands ranging from the visible spectrum to infrared wavelength [1]–[6]. However, in spite of a large number of HSIs spectral bands provide the possibility for subtly detecting the earth objects, they also introduce the high dimensionality, which brings several problems, e.g., huges phenomena [7], [8], heavy computational burden, and huge storage cost [9], [10]. In order to overcome this drawback, some methods, like the dimensionality reduction (DR) technique, have to be adopted in the processing of HSIs [10].

DR can be divided into feature extraction and feature selection [11]. Feature extraction achieves DR by finding a mapping function to project high-dimensional data into low-dimensional space [10], [12], such as principal component analysis (PCA) [13], nonnegative matrix factorization (NMF) [14], independent component analysis (ICA) [15], locality preserving projection (LPP) [16], and locally linear embedding (LLE) [17]. Feature selection is also called band selection (BS) in the HSIs domain, which selects a subset of the original features to reduce the dimensionality. Compared with feature extraction, BS methods select a subset from the original bands without losing their physical meaning and have the advantage of providing better interpretability for some geology analysis [10], [12]. Therefore, we focus on the BS approach in this article.

According to how much labeled information is used, BS models can be classified into supervised, unsupervised, and semisupervised. Supervised BS methods use labeled data to measure the importance of bands, such as mutual information (MI) [18], minimum estimation abundance covariance (MEAC) [19], discriminative kernel clustering [20], and nonhomogeneous hidden Markov chain (NHMC) [21]. However, the prior knowledge of training samples with a class label is usually scarce in most situations and labeling abundant training data is too expensive [12], [22]–[24], which limits the development of supervised methods.

On the contrary, unsupervised methods do not need to use training data to construct model, such as maximum variance principal component analysis (MVPCA) [25], information divergence (ID) [26], improved sparse subspace clustering (ISSC) [27], dual clustering [28], and fast and robust principal component analysis on Laplacian graph (FRPCALG) [8]. Unsupervised BS methods have important practical significance as they do not need any prior information. However, without any labeled information, the performance of the unsupervised BS model cannot achieve the best.

Manuscript received January 26, 2020; revised May 25, 2020 and September 8, 2020; accepted October 27, 2020. Date of publication November 30, 2020; date of current version November 24, 2021. This work was supported in part by The National Key Research and Development Program of China under Grant 2018YFB1107400 and in part by The National Natural Science Foundation of China under Grant 61871470, Grant 61751202, and Grant 61772427. *(Corresponding author: Feiping Nie.)*

Fang He, Weimin Jia, and Fenggan Zhang are with the Xi'an Research Institute of Hi-Tech, Xi'an 710025, China, and also with the Center for OPTical IMagery Analysis and Learning (OPTIMAL), Northwestern Polytechnical University, Xi'an 710072, China (e-mail: fanghe1107@gmail.com; jwm602@163.com; zfg417@163.com).

Feiping Nie and Xuelong Li are with the School of Computer Science, Center for OPTical IMagery Analysis and Learning (OPTIMAL), Northwestern Polytechnical University, Xi'an 710072, China (e-mail: feipingnie@gmail.com; xuelong_li@nwpu.edu.cn).

Rong Wang is with the Center for OPTical IMagery Analysis and Learning (OPTIMAL), Northwestern Polytechnical University, Xi'an 710072, China (e-mail: wangrong07@tsinghua.org.cn).

Digital Object Identifier 10.1109/TGRS.2020.3037746

1558-0644 © 2020 IEEE. Personal use is permitted, but republication/redistribution requires IEEE permission.
See https://www.ieee.org/publications/rights/index.html for more information.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 11:53:01 UTC from IEEE Xplore. Restrictions apply.

Consequently, the semisupervised BS method, which utilizes the limited labeled data and the plentiful unlabeled data to select the discriminative and informative feature subset, can alleviate the above problems. In this article, we focus on the semisupervised BS method.

Recently, many semisupervised BS methods attract a lot of attention. Chen *et al.* [29] combined Fisher's criteria and Graph Laplacian (FCGL) to construct a semisupervised BS model to improve HSIs classification accuracy. Su *et al.* [30] put forward semisupervised band clustering model to complete BS. Feng *et al.* [31] proposed semisupervised trivariate mutual information and graph regulation (STMIGR) by using a graph regulation term to select highly information features. Then, Feng *et al.* [12] introduced a semisupervised criterion to select highly discriminative and informative, and lowly redundant bands. Bai *et al.* [32] presented a hypergraph model that generalizes the traditional graph for BS. Cao *et al.* [33] proposed a semisupervised wrapper BS method, the BS based on dynamic classifier selection, to select the most discriminating bands.

Inspired by these works, we propose a semisupervised band selection with an optimal graph (BSOG) model for HSIs classification. The main contributions of our work are as follows.

1) A novel graph-based semisupervised BS model is proposed, which can not only perform BS but also learn local structure simultaneously. The local manifold structure is learned adaptively, which is helpful to select more valuable bands.
2) Instead of fixing the similarity matrix, the similarity matrix is constantly optimized to make the adjacent graph reliable, which is important for constructing the graph model. The model is more accurate with the better quality of the adjacent graph.

The rest of this article is organized as follows. In Section II, we have a brief survey on the related work of the BS model. In Section III, we propose a novel semisupervised BS model with an optimal graph. In Section IV, we adopt an alternative iterative algorithm to tackle the optimal problem. In Section V, we validate our methods and make comparisons with other approaches on four real HSIs data sets. In Section VI, we come to a conclusion and discuss the future work.

## II. Related Work

Sun and Du [34] reviewed and analyzed hyperspectral BS approaches in detail. They summarized that current hyperspectral BS methods can be classified into six main categories: ranking-based method, searching-based method, clustering-based method, sparsity-based method, embedding-learning-based method, and hybrid-scheme-based method.

The ranking-based approaches utilize a predefined band-prioritization criterion to quantify the importance of each spectral band and select the top-ranked bands. The typical methods are minimum-variance PCA (MVPCA) [25], enhanced fast density peak-based clustering (E-FDPC) [35], NHMC [21], and so on. The searching-based methods, like linear prediction (LP) [36] and MEAC [19], convert BS into an optimization problem of a given criterion function and select the best bands by solving this optimization problem. The clustering-based methods select representative bands from band clusters, which can be divided into: $k$-means based [28], affinity propagation (AP)-based [37], and graph-based method [38]. The sparsity-based BS methods reveal certain underlying structures within HSI data by sparse representation or sparse regression. Then, the BS problem becomes solving an optimization problem with sparsity constraints. The representative sparsity-constrained methods include sparse nonnegative matrix factorization (SNMF)-based approaches [39], sparse representation-based approaches [27], and sparse regression-based approaches [40]. The embedding-learning-based methods combine BS with the optimization of specific application models, e.g., embedded feature selection (EFS) [41], sparse multinomial linear regression (SMLR) [42], sparsity-promoting priors [43], deep convolutional neural networks, and distance density [44]. The hybrid-scheme-based methods integrate two or more different schemes into BS, e.g., combined ranking and clustering criteria [45], combined ranking, and searching criteria [46].

Besides, according to whether utilizes the prior information, all of these BS methods can be divided into supervised approaches, unsupervised approaches, and semisupervised approaches. After an in-depth analysis, Sun and Du [34] pointed out that semisupervised BS methods have more important practical significance and should be developed to promote BS performance. In this article, we focus on the graph-based semisupervised BS model.

Next, we take the FCGL model as an example to review the graph-based semisupervised BS method. First, the notations and definitions referred to in this article are summarized. Then, FCGL model [29] is introduced. Finally, we give a brief analysis of the FCGL algorithm.

### A. Notations and Definitions

In this article, matrices and vectors are written as boldface uppercase and lowercase letters, respectively. For example, a matrix $\boldsymbol{M} \in \mathbb{R}^{n\times m}$, its $i$th row is denoted $\boldsymbol{m}^i$, and its $j$th column is $\boldsymbol{m}_j$. The $\ell_p$-norm of the vector $\boldsymbol{v} \in \mathbb{R}^n$ is defined as $\|\boldsymbol{v}\|_p = (\sum_{i=1}^{n} |v_i|^p)^{(1/p)}$. The Frobenius norm of the matrix $\boldsymbol{M} \in \mathbb{R}^{n\times m}$ is

$$\|\boldsymbol{M}\|_F = \sqrt{\sum_{i=1}^{n}\sum_{j=1}^{m} m_{ij}^2} = \sqrt{\sum_{i=1}^{n} \|\boldsymbol{m}^i\|_2^2}. \tag{1}$$

Then $\ell_{2,1}$-norm of the matrix $\boldsymbol{M}$ is defined as

$$\|\boldsymbol{M}\|_{2,1} = \sum_{i=1}^{n} \sqrt{\sum_{j=1}^{m} m_{ij}^2} = \sum_{i=1}^{n} \|\boldsymbol{m}^i\|_2. \tag{2}$$

In semisupervised situation, the HSI data matrix $\boldsymbol{X} = [\boldsymbol{x}_1, \ldots, \boldsymbol{x}_l, \boldsymbol{x}_{l+1}, \ldots, \boldsymbol{x}_n]^{\mathrm{T}} \in \mathbb{R}^{n\times d}$ usually represents the first $l$ samples labeled, while the remaining are unlabeled, where $n$ and $d$ are the number of total data points and bands, respectively. Let $\Phi = \{\boldsymbol{\phi}_1, \ldots, \boldsymbol{\phi}_d\}$ denote the set of $d$ features. Then, $\phi_{ri}$ is the $i$th sample of the $r$th feature, $i = 1, \ldots, n$.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 11:53:01 UTC from IEEE Xplore. Restrictions apply.

Then, each data point $\boldsymbol{x}_i$ can be viewed as a vertex of $V$ in the affinity graph $G = (V, E)$, where $E$ is the set of edges. Let $w_{ij}$ represent the weight of the edge between $\boldsymbol{x}_i$ and $\boldsymbol{x}_j$, which measures the similarity relationship of them. Therefore, the similarity matrix of the affinity graph is $\boldsymbol{W} = \{w_{ij}\} \in \mathbb{R}^{n\times n}$, $\forall i, j \in \{1, \ldots, n\}$.

Let $\boldsymbol{Y} = [\boldsymbol{y}_1, \ldots, \boldsymbol{y}_l, \boldsymbol{y}_{l+1}, \ldots, \boldsymbol{y}_n]^{\mathrm{T}} \in \mathbb{R}^{n\times c}$ represent the label matrix, where $c$ is the number of classes. If the $i$th data labeled as $j$, $y_{ij} = 1$. Otherwise, $y_{ij} = 0$. $\boldsymbol{F} = [\boldsymbol{f}_1, \ldots, \boldsymbol{f}_l, \boldsymbol{f}_{l+1}, \ldots, \boldsymbol{f}_n]^{\mathrm{T}} \in \mathbb{R}^{n\times c}$ is the corresponding soft label matrix.

### B. FCGL Model

Next, we review the FCGL model [29]. FCGL method achieved BS by constructing the following within-class affinity matrix $\boldsymbol{S}_w$ and between-class affinity matrix $\boldsymbol{S}_b$:

$$S_{w,ij} = \begin{cases} \zeta, & \text{if } \boldsymbol{x}_i \text{ and } \boldsymbol{x}_j \text{ share the same label} \\ 1, & \text{if } \boldsymbol{x}_i \text{ or } \boldsymbol{x}_j \text{ is unlabeled, but} \\ & \boldsymbol{x}_i \in \mathrm{KNN}(j) \text{ or } \boldsymbol{x}_j \in \mathrm{KNN}(i) \\ 0, & \text{otherwise} \end{cases} \tag{3}$$

where $k$-nearest neighbor (KNN) $(i)$ is the set of $k$ neighbors of node $i$. Similarly, KNN$(j)$ is the set of $k$ neighbors of node $j$. $\zeta$ is a suitable constant

$$S_{b,ij} = \begin{cases} 1, & \text{if } \boldsymbol{x}_i \text{ and } \boldsymbol{x}_j \text{ belongs to} \\ & \text{the different classes} \\ 0, & \text{otherwise.} \end{cases} \tag{4}$$

Based on $\boldsymbol{S}_w$ and $\boldsymbol{S}_b$, the graph Laplacians of $\boldsymbol{G}_w$ and $\boldsymbol{G}_b$ can be computed as follows:

$$\begin{aligned} \boldsymbol{L}_w &= \boldsymbol{D}_w - \boldsymbol{S}_w, \quad \boldsymbol{D}_w = \mathrm{diag}(\boldsymbol{S}_w\mathbf{1}) \\ \boldsymbol{L}_b &= \boldsymbol{D}_b - \boldsymbol{S}_b, \quad \boldsymbol{D}_b = \mathrm{diag}(\boldsymbol{S}_b\mathbf{1}) \end{aligned} \tag{5}$$

where $\mathbf{1} = (1, 1, \ldots, 1)^{\mathrm{T}}$.

Then, FCGL introduces the following loading factor [25]:

$$\boldsymbol{O}_{\delta i} = \sqrt{r_i}\, p_{\delta i}, \quad \delta = 1, 2, \ldots, d \tag{6}$$

where $\boldsymbol{p}_i = [p_{1i}, p_{2i}, \ldots, p_{di}]^T$ is an eigenvector, $r_i (1 \le i \le d)$ is the corresponding eigenvalue. $\boldsymbol{P} = [\boldsymbol{p}_1, \boldsymbol{p}_2, \ldots, \boldsymbol{p}_n]^{\mathrm{T}}$ is a linear transformation matrix. The optimal $\boldsymbol{P}$ is given by solving the following generalized eigenvalue problem as follows:

$$\left(\boldsymbol{X}\boldsymbol{L}_b\boldsymbol{X}^T\right)\boldsymbol{P} = \left(\boldsymbol{X}\boldsymbol{L}_w\boldsymbol{X}^T\right)\boldsymbol{P}\boldsymbol{R} \tag{7}$$

where $\boldsymbol{R} = \mathrm{diag}(r_1, r_2, \ldots, r_d)$ is a diagonal matrix.

Then, the weight for each feature is defined by

$$w(\delta) = \sum_{i=1}^{\delta} O_{\delta i}^2. \tag{8}$$

According to (8), each feature can be ranked based on its weight. The feature with a higher weight is to be selected into the feature subset. The detailed step of the FCGL model is described in Algorithm 1.

The corresponding main computational complexity of FCGL is $O(3n^2d + 2nd^2 + n^3)$.

**Algorithm 1** Algorithm of FCGL

**Input:** data matrix $\boldsymbol{X} \in \mathbb{R}^{n\times d}$, the number of selecting band $m$ and nearest neighbor $k$, the parameter $\zeta$.

Step1: construct the within-class affinity matrix $\boldsymbol{S}_w$ by (3) and between-class affinity matrix $\boldsymbol{S}_b$ by (4).

Step2: calculate the corresponding Laplacian matrix of $\boldsymbol{S}_w$ and $\boldsymbol{S}_b$ by (5).

Step3: calculate the eigenvector $p_i$ by (7).

Step4: calculate the loading factor by (6).

Step5: calculate the weight for each feature by (8).

**Output:** calculate all feature weight and sort it in descending order, select top $m$ ranked bands as ultimate result.

### C. Analysis of FCGL Model

Overall, the FCGL model is based on manifold learning and spectral graph theory, which tries to discover both geometrical and discriminant structure in the data by constructing two graphs, i.e., within-class graph and between-class graph [29], [47].

However, FCGL still exhibits two limitations. On the one hand, the two graphs of them are calculated from the original data and stay the same. In the real-world, HSIs often contain a large number of features and samples with noise, which has a bad effect on the quality of the graph. On the other hand, it ranks the bands by calculating the feature's weight, while ignores the characteristics of the model itself. That is, FCGL performs BS and constructs graphs independently.

## III. Semisupervised BS With Graph Optimization

In this section, we propose a novel semisupervised BS model based on a structured graph. In order to get a high-quality graph, a learned way is employed to update the graph other than fixing the initial graph. In addition, feature selection and local structure graph optimizing are performed simultaneously to achieve a better result.

### A. Initialization Structured Graph

First, we need to give an initial graph. In this article, we adopt the following adaptive process to construct the similarity matrix with probabilistic neighbors [48]:

$$\min_{s_i^T\mathbf{1}=1, s_{ij}\ge 0} \sum_{i,j=1}^{n} \left\|\boldsymbol{x}_i - \boldsymbol{x}_j\right\|_2^2 s_{ij} + \alpha s_{ij}^2 \tag{9}$$

where $\boldsymbol{s}_i^T$ denotes the $i$th row of $\boldsymbol{S}$. $\boldsymbol{S} \in \mathbb{R}^{n\times n}$ is the similarity matrix. $\alpha$ is the regularization parameter. The regularization term is used to avoid the trivial solution when the nearest data point can be the neighbor of $\boldsymbol{x}_i$ with probability 1 and all the other data points cannot be the neighbors of $\boldsymbol{x}_i$ [49].

Denote $z_{ij}^x = \|\boldsymbol{x}_i - \boldsymbol{x}_j\|_2^2$, and $\boldsymbol{z}_i^x \in \mathbb{R}^{n\times 1}$ as a vector with $j$th element as $z_{ij}^x$. Then, (9) can be rewritten in the following vector form:

$$\min_{s_i^T\mathbf{1}=1, \boldsymbol{s}_i\ge 0} \left\|\boldsymbol{s}_i + \frac{1}{2\alpha}\boldsymbol{z}_i^x\right\|_2^2. \tag{10}$$

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 11:53:01 UTC from IEEE Xplore. Restrictions apply.

The Lagrangian function of (10) is

$$\mathcal{L}(\boldsymbol{s}_i, \eta, \boldsymbol{\beta}_i) = \frac{1}{2}\left\|\boldsymbol{s}_i + \frac{1}{2\alpha}\boldsymbol{z}_i^x\right\|_2^2 - \eta\left(\boldsymbol{s}_i^T\mathbf{1} - 1\right) - \boldsymbol{\beta}_i^T\boldsymbol{s}_i \quad (11)$$

where $\eta$ and $\beta_i \geq 0$ are the Lagrangian multipliers.

According to the Karush–Kuhn–Tucker (KKT) condition, it can be verified that the optimal solution $\boldsymbol{s}_i$ should be

$$s_{ij} = \left(-\frac{z_{ij}^x}{2\alpha} + \beta\right)_+ \quad (12)$$

where $\beta = (1/k) + 1/(2k\alpha_i)\sum_{j=1}^{k} z_{ij}$ [49] and $k$ is the number of nearest neighbors. For detailed derivation, see [48]. According to [48], the learned $\boldsymbol{S}$ is naturally sparse, scale-invariant and only involves the basic operations of addition, subtraction, multiplication, and division.

In practice, if we focus on the locality of data, we could achieve better performance. Therefore, it is preferred to learn a sparse $\boldsymbol{s}_i$, i.e., only the $k$ nearest neighbors of $\boldsymbol{x}_i$ could have chance to connect to $\boldsymbol{x}_i$. Besides, learning a sparse similarity matrix $\boldsymbol{S}$ can alleviate the computation burden to a large extent. To make $\boldsymbol{s}_i$ exactly contain $k$ nonzero elements, $\boldsymbol{s}_i$ must satisfy $s_{i,k+1} \leq 0 < s_{i,k}$, thus $\alpha_i$ should be set as

$$\frac{k}{2}z_{ik} - \frac{1}{2}\sum_{j=1}^{k} z_{ij} < \alpha_i \leq \frac{k}{2}z_{i,k+1} - \frac{1}{2}\sum_{j=1}^{k} z_{ij} \quad (13)$$

where $z_{i1}, z_{i2}, \ldots, z_{\text{in}}$ are sorted in ascending order. To make most of $\boldsymbol{s}_i$ has $k$ nonzeros elements, $\alpha$ could be set as

$$\alpha = \frac{1}{n}\sum_{i=1}^{n}\alpha_i = \frac{1}{n}\sum_{i=1}^{n}\left(\frac{k}{2}z_{i,k+1} - \frac{1}{2}\sum_{j=1}^{k} z_{ij}\right). \quad (14)$$

### *B. Semisupervised BS With Graph Optimization*

Once the initial structured graph is obtained, according to the theory of manifold learning, a linear combination $\boldsymbol{XW}$ of original features to best approximate the low-dimensional manifold can be found. $\boldsymbol{W} \in \mathbb{R}^{d\times m}$ is the projection matrix, where $d$ and $m$ are the original dimension and projected dimension, respectively.

Nie *et al.* [50] proposed an unsupervised feature selection with structured graph optimization model, the goal of which is to solve the following problem:

$$\begin{aligned}\min \sum_{i,j=1}^{n}\left(\left\|\boldsymbol{W}^T\boldsymbol{x}_i - \boldsymbol{W}^T\boldsymbol{x}_j\right\|_2^2 s_{ij} + \alpha_u s_{ij}^2\right) \\ + \gamma_u\|\boldsymbol{W}\|_{2,1} + 2\lambda_u \mathrm{Tr}\left(\boldsymbol{F}^T\boldsymbol{L}_S\boldsymbol{F}\right) \\ \text{s.t. } \forall i, \quad \boldsymbol{s}_i^T\mathbf{1} = 1,\ s_{ij} \geq 0, \\ \boldsymbol{F} \in \mathbb{R}^{n\times c},\ \boldsymbol{F}^T\boldsymbol{F} = \boldsymbol{I},\ \boldsymbol{W}^T\boldsymbol{W} = \boldsymbol{I}\end{aligned} \quad (15)$$

where $\alpha_u$ and $\gamma_u$ are the regularization parameters. $\lambda_u$ is large enough.

Inspired by this work, we propose the following BSOG model:

$$\begin{aligned}\min \sum_{i,j=1}^{n}\left(\left\|\boldsymbol{W}^T\boldsymbol{x}_i - \boldsymbol{W}^T\boldsymbol{x}_j\right\|_2^2 s_{ij} + \alpha s_{ij}^2\right) \\ + \gamma\|\boldsymbol{W}\|_{2,1} + \lambda\mathrm{Tr}\left(\boldsymbol{F}^T\boldsymbol{L}_S\boldsymbol{F}\right) \\ \text{s.t. } \forall i, \quad \boldsymbol{s}_i^T\mathbf{1} = 1,\ s_{ij} \geq 0,\ \boldsymbol{W}^T\boldsymbol{W} = \boldsymbol{I},\ \boldsymbol{F}_l = \boldsymbol{Y}_l\end{aligned} \quad (16)$$

where $\alpha$, $\gamma$, and $\lambda$ are the regularization parameters.

In a semisupervised situation, we hope that a good classification function should not change too much on the difference between the resulted labels and the assignment of initial labels. Therefore, $\boldsymbol{F}_l = \boldsymbol{Y}_l$ is added to be a constraint.

$\|\boldsymbol{W}^T\boldsymbol{x}_i - \boldsymbol{W}^T\boldsymbol{x}_j\|_2^2\, s_{ij}$ is a smoothness term, measuring the smoothness of the projected features, meaning that the closer data points in the high-dimensional space still maintain a close distance in the low dimensional space after projected.

$\alpha s_{ij}^2$ is a regularization term, which is adopted to avoid the trivial solution.

$\gamma\|\boldsymbol{W}\|_{2,1}$ is a sparsity regularization term. The sparsity regularization on $\boldsymbol{W}$ makes it suitable for selecting valuable features. Using the $\ell_{2,1}$-norm constraint to complete feature selection has been proven effective and investigated widely among sparse learning-based feature selection field [51]–[53]. Minimizing $\boldsymbol{L}_{2,1}$ can help to learn a row sparse projection matrix $\boldsymbol{W}$ in which only a few rows of $\boldsymbol{W}$ are nonzeros. With this row sparse projection matrix $\boldsymbol{W}$, only a few important features are involved in the projection. Besides, only those discriminative features would have nonzero weights in $\boldsymbol{W}$, and thus each new projected feature is a linear combination of these discriminative features only. In this way, only discriminative information is retained. More details can be found in [51] and [54]. Therefore, in this article, we also adopt $\ell_{2,1}$-norm regularization on $\boldsymbol{W}$ to make it row sparse. To make the feature space distinctive after reduction, following Nie *et al.* [54] and Wang *et al.* [50], we use the constraint $\boldsymbol{W}^T\boldsymbol{W} = \boldsymbol{I}$ to replace regular $\boldsymbol{W}^T\boldsymbol{S}_t\boldsymbol{W} = \boldsymbol{I}$, where $\boldsymbol{S}_t$ is the covariance matrix of data $\boldsymbol{X}$.

Then, the importance of the $i$th feature can be measured by $\|w_i\|_2$. The most important $m$ bands are selected by the sorted $\|w_i\|_2$, where $m$ is the number of bands that need to be selected.

In (16), $\boldsymbol{W}$ is utilized to select features and $\boldsymbol{S}$ is used to learn local structure. Therefore, our BSOG model can perform BS and local structure learning simultaneously.

## IV. Optimization Algorithm

Equation (16) contains three different variables and $\ell_{2,1}$-norm regularization, which makes it hard to solve. Therefore, we adopt the following alternative iterative algorithm to tackle (16).

### *A. Fix $\boldsymbol{S}$, Update $\boldsymbol{W}$*

First, we fix $\boldsymbol{S}$. Then, (16) can be turned into

$$\min_{\boldsymbol{W}^T\boldsymbol{W}=\boldsymbol{I}} \sum_{i,j=1}^{n}\left\|\boldsymbol{W}^T\boldsymbol{x}_i - \boldsymbol{W}^T\boldsymbol{x}_j\right\|_2^2 s_{ij} + \gamma\|\boldsymbol{W}\|_{2,1}. \quad (17)$$

According to the basic spectral analysis equation

$$\sum_{i,j=1}^{n}\left\|\boldsymbol{f}_i - \boldsymbol{f}_j\right\|_2^2 s_{ij} = 2\mathrm{Tr}\left(\boldsymbol{F}^T\boldsymbol{L}_S\boldsymbol{F}\right) \quad (18)$$

where $\boldsymbol{L}_S = \boldsymbol{D} - (\boldsymbol{S}^T + \boldsymbol{S})/2$ is Laplacian matrix and $\boldsymbol{D}$ is a diagonal matrix with $i$th entry being $\sum_j (s_{ij} + s_{ji})/2$.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 11:53:01 UTC from IEEE Xplore. Restrictions apply.

**Algorithm 2** Algorithm of Solving (16)

**Input:** training data matrix $\boldsymbol{X} \in \mathbb{R}^{n\times d}$, Laplacian Matrix $\boldsymbol{L}_S \in \mathbb{R}^{n\times n}$, the number of selecting bands $m$, the regularization parameter $\gamma$.

Step1: initialize $\boldsymbol{Q} \in \mathbb{R}^{d\times d}$, $\boldsymbol{Q} = \boldsymbol{I}$.

Step2: with current $\boldsymbol{Q}$, calculating $\boldsymbol{W}$ by solving (24).

Step3: with current $\boldsymbol{W}$, calculating $\boldsymbol{Q}$ by solving (22).

Step4: **repeat** Step2 and Step3 **until** converge.

**Output:** projection matrix $\boldsymbol{W} \in \mathbb{R}^{d\times m}$.

Replace $\|\boldsymbol{W}\|_{2,1}$ with $\sum_i \|\boldsymbol{w}_i\|_2$. Equation (16) can be transformed into

$$\min_{\boldsymbol{W}^T\boldsymbol{W}=\boldsymbol{I}} \mathrm{Tr}\big(\boldsymbol{W}^T\boldsymbol{X}^T\boldsymbol{L}_S\boldsymbol{X}\boldsymbol{W}\big) + \gamma \sum_i \|\boldsymbol{w}_i\|_2. \tag{19}$$

In theory, $\|\boldsymbol{w}_i\|_2$ can be zero, which will make (19) nondifferentiable. To avoid this situation, $\|\boldsymbol{w}_i\|_2$ will be replaced as $(\boldsymbol{w}_i^T\boldsymbol{w}_i + \varepsilon)^{1/2}$, where $\varepsilon$ is a small enough constant. Then (19) becomes

$$\min_{\boldsymbol{W}^T\boldsymbol{W}=\boldsymbol{I}} \mathrm{Tr}\big(\boldsymbol{W}^T\boldsymbol{X}^T\boldsymbol{L}_S\boldsymbol{X}\boldsymbol{W}\big) + \gamma \sum_i \sqrt{\boldsymbol{w}_i^T\boldsymbol{w}_i + \varepsilon}. \tag{20}$$

To solve (20), we apply the following Lagrangian function of (20):

$$\mathcal{L}(\boldsymbol{W}, \boldsymbol{\Delta}) = \mathrm{Tr}\big(\boldsymbol{W}^T\boldsymbol{X}^T\boldsymbol{L}_S\boldsymbol{X}\boldsymbol{W}\big) + \gamma \sum_i \sqrt{\boldsymbol{w}_i^T\boldsymbol{w}_i + \varepsilon} + \mathrm{Tr}\big(\boldsymbol{\Delta}\big(\boldsymbol{W}^T\boldsymbol{W} - \boldsymbol{I}\big)\big) \tag{21}$$

where $\boldsymbol{\Delta}$ is the Lagrangian multiplier. Setting the derivative of $\mathcal{L}(\boldsymbol{W}, \boldsymbol{\Delta})$ to zero, we have

$$\frac{\partial \mathcal{L}(\boldsymbol{W}, \boldsymbol{\Delta})}{\partial \boldsymbol{W}} = \boldsymbol{X}^T\boldsymbol{L}_S\boldsymbol{X}\boldsymbol{W} + \gamma\,\boldsymbol{Q}\boldsymbol{W} + \boldsymbol{W}\boldsymbol{\Delta} = 0 \tag{22}$$

where $\boldsymbol{Q} \in \mathbb{R}^{d\times d}$ with $i$th element defined as

$$Q_{ii} = \frac{1}{2\sqrt{\boldsymbol{w}_i^T\boldsymbol{w}_i + \varepsilon}}. \tag{23}$$

From (23), we can find that $\boldsymbol{Q}$ is also a variable, which depends on $\boldsymbol{W}$. Therefore, we also adopt an iterative algorithm to solve (20). When $\boldsymbol{W}$ is fixed, $\boldsymbol{Q}$ can be calculated by (23). When $\boldsymbol{Q}$ is fixed, it can be easily proved that solving (22) is equivalent to solving

$$\min_{\boldsymbol{W}^T\boldsymbol{W}=\boldsymbol{I}} \mathrm{Tr}\big(\boldsymbol{W}^T\boldsymbol{X}^T\boldsymbol{L}_S\boldsymbol{X}\boldsymbol{W}\big) + \gamma\,\mathrm{Tr}\big(\boldsymbol{W}^T\boldsymbol{Q}\boldsymbol{W}\big). \tag{24}$$

Therefore, the optimal solution $\boldsymbol{W}$ is formed by the $m$ eigenvectors of $(\boldsymbol{X}^T\boldsymbol{L}_S\boldsymbol{X} + \gamma\,\boldsymbol{Q})$ corresponding to the $m$ smallest eigenvalues. The detailed step of solving (17) is summarized in Algorithm 2.

### *B. Fix $\boldsymbol{S}$, $\boldsymbol{W}$, Update $\boldsymbol{F}$*

With fixed $\boldsymbol{S}$, (16) can be transformed into

$$\mathcal{J} = \min_{\boldsymbol{F}_l=\boldsymbol{Y}_l} \mathrm{Tr}\big(\boldsymbol{F}^T\boldsymbol{L}_S\boldsymbol{F}\big) \tag{25}$$

where $\boldsymbol{F}$ and $\boldsymbol{L}_S$ can be rewritten as

$$\boldsymbol{F} = \begin{bmatrix}\boldsymbol{F}_l\\ \boldsymbol{F}_u\end{bmatrix}, \quad \boldsymbol{L}_S = \begin{bmatrix}\boldsymbol{L}_{ll} & \boldsymbol{L}_{lu}\\ \boldsymbol{L}_{ul} & \boldsymbol{L}_{uu}\end{bmatrix},$$

respectively. Then, (25) becomes

$$\mathcal{J} = \min_{\boldsymbol{F}_l=\boldsymbol{Y}_l} \mathrm{Tr}\left(\begin{bmatrix}\boldsymbol{F}_l\\ \boldsymbol{F}_u\end{bmatrix}^T\begin{bmatrix}\boldsymbol{L}_{ll} & \boldsymbol{L}_{lu}\\ \boldsymbol{L}_{ul} & \boldsymbol{L}_{uu}\end{bmatrix}\begin{bmatrix}\boldsymbol{F}_l\\ \boldsymbol{F}_u\end{bmatrix}\right). \tag{26}$$

The optimal solution can be easily solved by

$$\frac{\partial \mathcal{J}}{\partial \boldsymbol{F}_u} = \boldsymbol{L}_{ul}\boldsymbol{F}_l + \boldsymbol{F}_l^T\boldsymbol{L}_{lu} + 2\boldsymbol{L}_{uu}\boldsymbol{F}_u = 0. \tag{27}$$

Then, the final solution is

$$\boldsymbol{F}_u = -\boldsymbol{L}_{uu}^{-1}\boldsymbol{L}_{ul}\boldsymbol{Y}_l. \tag{28}$$

### *C. Fix $\boldsymbol{W}$, $\boldsymbol{F}$, Update $\boldsymbol{S}$*

With fixed $\boldsymbol{W}$ and $\boldsymbol{F}$, (16) can be transformed into

$$\begin{aligned}\min\ & \sum_{i,j=1}^{n}\Big(\big\|\boldsymbol{W}^T\boldsymbol{x}_i - \boldsymbol{W}^T\boldsymbol{x}_j\big\|_2^2 s_{ij} + \alpha s_{ij}^2\Big)\\ & + \lambda\,\mathrm{Tr}\big(\boldsymbol{F}^T\boldsymbol{L}_S\boldsymbol{F}\big)\\ \text{s.t.}\ & \ \forall i, \quad \boldsymbol{s}_i^T\boldsymbol{1} = 1,\ s_{ij} \ge 0.\end{aligned} \tag{29}$$

According to (18), (29) is equivalent to

$$\begin{aligned}\min\ & \sum_{i,j=1}^{n}\Big(\big\|\boldsymbol{W}^T\boldsymbol{x}_i - \boldsymbol{W}^T\boldsymbol{x}_j\big\|_2^2 s_{ij} + \alpha s_{ij}^2\Big)\\ & + \frac{\lambda}{2}\sum_{i,j=1}^{n}\big\|\boldsymbol{f}_i - \boldsymbol{f}_j\big\|_2^2 s_{ij}\\ \text{s.t.}\ & \ \forall i, \quad \boldsymbol{s}_i^T\boldsymbol{1} = 1,\ s_{ij} \ge 0.\end{aligned} \tag{30}$$

As the similarity vector of each sample is independent, for the $i$th sample, the following problem needs to be solved:

$$\begin{aligned}\min\ & \sum_{j=1}^{n}\Big(\big\|\boldsymbol{W}^T\boldsymbol{x}_i - \boldsymbol{W}^T\boldsymbol{x}_j\big\|_2^2 s_{ij} + \alpha s_{ij}^2\Big)\\ & + \frac{\lambda}{2}\sum_{j=1}^{n}\big\|\boldsymbol{f}_i - \boldsymbol{f}_j\big\|_2^2 s_{ij}\\ \text{s.t.}\ & \ \forall i, \quad \boldsymbol{s}_i^T\boldsymbol{1} = 1,\ s_{ij} \ge 0.\end{aligned} \tag{31}$$

Let $p_{ij} = \|\boldsymbol{W}^T\boldsymbol{x}_i - \boldsymbol{W}^T\boldsymbol{x}_j\|_2^2$, $\boldsymbol{P} \in \mathbb{R}^{n\times n}$, $n_{ij} = \|\boldsymbol{f}_i - \boldsymbol{f}_j\|_2^2$, $\boldsymbol{N} \in \mathbb{R}^{n\times n}$. Define vector $\boldsymbol{h}_i \in \mathbb{R}^{n\times 1}$ with $h_{ij} = p_{ij} + \frac{\lambda}{2}n_{ij}$. Then, (31) can be simplified as

$$\min_{\boldsymbol{s}_i^T\boldsymbol{1}=1,\ \ s_{ij}\ge 0}\left\|\boldsymbol{s}_i + \frac{1}{2\alpha}\boldsymbol{h}_i\right\|_2^2. \tag{32}$$

Note that the form of (32) is the same as (10), thus the solution is also similar to (12), that is

$$s_{ij} = \left(-\frac{h_{ij}}{2\alpha} + \beta\right)_+ \tag{33}$$

where $\alpha = (k/2)h_{i,k+1} - (1/2)\sum_{j=1}^{k} h_{ij}$, $\beta = (1/k) + 1/(2k\alpha_i)\sum_{j=1}^{k} h_{ij}$.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 11:53:01 UTC from IEEE Xplore. Restrictions apply.

**Algorithm 3** Algorithm of BSOG

**Input:** training data matrix $\boldsymbol{X} \in \mathbb{R}^{n\times d}$, which is consisting of labeled data $\boldsymbol{X}_l \in \mathbb{R}^{l\times d}$ and unlabeled data $\boldsymbol{X}_u \in \mathbb{R}^{u\times d}$, the number of selecting band $m$ and nearest neighbor $k$, the regularization parameter $\lambda$ and $\gamma$.

Step1: initialize $\boldsymbol{S}$ by (12), which needs $O(n^2d)$.

Step2: update $\boldsymbol{W}$ via Algorithm 2, which needs $O(2d^2m + d^2n)$.

Step3: calculate $\boldsymbol{L}_S$ and rewrite $\boldsymbol{L}_S$ as $\boldsymbol{L}_S = \begin{bmatrix} \boldsymbol{L}_{ll} & \boldsymbol{L}_{lu} \\ \boldsymbol{L}_{ul} & \boldsymbol{L}_{uu} \end{bmatrix}$.

Step4: update $\boldsymbol{F}_u$ by (28), which needs $O(u^3 + u^2c + ulc)$.

Step5: update $\boldsymbol{S}$ by (34).

**Output:** calculate all $\|\boldsymbol{w}_i\|_2 (i = 1, 2, \ldots, n)$ and sort in descending order, select top $m$ ranked bands as ultimate result.

The detailed solution of $s_{ij}$ is

$$s_{ij} = \begin{cases} \dfrac{h_{i,k+1} - h_{i,j}}{kh_{i,k+1} - \sum_{j=1}^{k} h_{i,j}}, & j \le k \\ 0, & j > k. \end{cases} \tag{34}$$

To summarize, we demonstrate the steps of BSOG in Algorithm 3.

The corresponding computational complexity of BSOG is $O(n^2d + 2d^2m + d^2n + u^3 + u^2c + ulc)$, where $t$, $l$, and $u$ are the number of iterations labeled and unlabeled samples, respectively.

## V. Experiments

In this section, several typical unsupervised and semisupervised BS methods are chosen as comparison methods to verify the effectiveness of the present method. For the BSOG method, all the parameters are adjusted to the optimal. For other methods used for comparison, the parameters are set according to their original text. After selecting suitable bands by these approaches, the support vector machine (SVM) is adopted to complete the classification task. The average classification accuracy (AA), the overall classification accuracy (OA), and the kappa coefficient ($\kappa$) are viewed as the classified quantitative evaluations to assess each algorithm. All our experiments are completed on a Windows 10 computer with a 3.6 GHz Intel i7-7700 CPU and 32 GB RAM, MATLAB 2017a.

### A. Data Sets Description

To evaluate the performance of our BSOG model, we conduct experiments on several real-world data sets. The detailed descriptions of all data sets are given as follows, and the RGB image and corresponding ground truth map are shown in Fig. 1.

1) *Indian Pines:* This image was acquired by the Airborne Visible Infrared Imaging Spectrometer (AVIRIS) sensor over the North-western Indian in 1992, with 145 × 145 pixels and 224 bands. After removing the noisy and water absorption bands, there remain 200 bands. There are 10 249 samples containing 16 classes available. The false color image which consists of bands 40, 30, and 20 is present in Fig. 1(a). The ground truth map is provided in Fig. 1(e).
2) *Salinas:* The data set was also acquired by the AVIRIS sensor over Salinas Valley, California, with 512 × 217 pixels and 224 bands. There are 204 bands and 54 129 samples containing 16 classes available. The false color image consisting of bands 70, 27, and 17, and the corresponding ground truth map are, respectively, provided in Fig. 1(b) and (f).
3) *Botswana:* The data set was acquired by the Hyperion sensor in the Okavango Delta region of Botswana in May 2001, with 1476 × 256 pixels. There are 145 bands and 3248 samples containing 14 classes available for experiment analysis. The false color image consists of bands 60, 25, and 16, and the corresponding ground-truth map are, respectively, provided in Fig. 1(c) and (g).

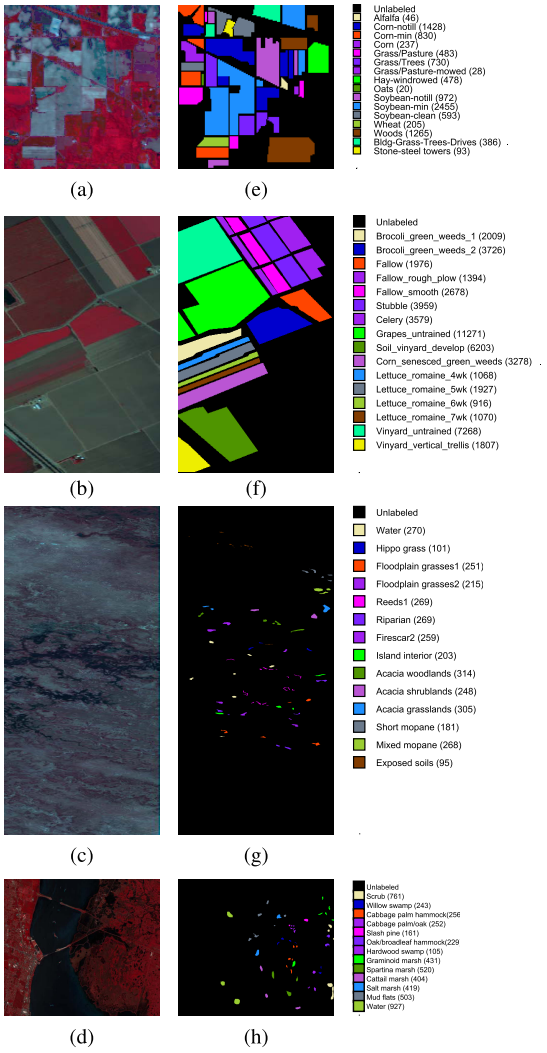



Fig. 1. False color images and ground truth maps of four data sets. (a) False color image of Indian Pines. (b) False color image of Salinas. (c) False color image of Botswana. (d) False color image of KSC. (e)–(h) Corresponding ground-truth maps. (Note that the number of each class is shown in brackets.)

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 11:53:01 UTC from IEEE Xplore. Restrictions apply.

TABLE I
CLASSIFICATION RESULTS ACHIEVED USING 80 SELECTED BANDS FROM THE INDIAN PINES DATA SET

| Class | Labeled | Unlabeled | Test | Samples | Baseline | ISSC | OCF | SOPSRL | ONR | FCGL | BSOG |
|---|---|---|---|---|---|---|---|---|---|---|---|
| Alfalfa | 5 | 9 | 32 | 46 | 0.6146 | 0.5833 | 0.7188 | 0.5104 | 0.5417 | 0.6458 | 0.8542 |
| Corn-notill | 143 | 286 | 999 | 1428 | 0.7971 | 0.7417 | 0.7534 | 0.7501 | 0.7314 | 0.7761 | 0.8408 |
| Corn-min | 83 | 166 | 581 | 830 | 0.6885 | 0.6701 | 0.6982 | 0.6517 | 0.6099 | 0.6695 | 0.8296 |
| Corn | 24 | 47 | 166 | 237 | 0.6345 | 0.5382 | 0.6165 | 0.5783 | 0.5522 | 0.6104 | 0.8233 |
| Grass/Pasture | 48 | 97 | 338 | 483 | 0.9152 | 0.9191 | 0.9034 | 0.8955 | 0.9063 | 0.9073 | 0.9181 |
| Grass/Trees | 73 | 146 | 511 | 730 | 0.9720 | 0.9667 | 0.9654 | 0.9524 | 0.96351 | 0.9589 | 0.9850 |
| Grass/Pasture-mowed | 5 | 5 | 18 | 28 | 0.8519 | 0.8333 | 0.7593 | 0.7407 | 0.8148 | 0.7407 | 0.8889 |
| Hay-windrowed | 48 | 96 | 334 | 478 | 0.9880 | 0.9890 | 0.9900 | 0.9850 | 0.9820 | 0.9920 | 0.9940 |
| Oats | 5 | 5 | 10 | 20 | 0.8667 | 0.9000 | 0.8333 | 0.8000 | 0.7667 | 0.8667 | 0.9667 |
| Soybean-notill | 97 | 194 | 681 | 972 | 0.8096 | 0.7734 | 0.7641 | 0.7651 | 0.7489 | 0.7910 | 0.8835 |
| Soybean-min | 246 | 491 | 1718 | 2455 | 0.8653 | 0.8335 | 0.8438 | 0.8236 | 0.8378 | 0.8539 | 0.8861 |
| Soybean-clean | 59 | 119 | 415 | 593 | 0.7422 | 0.6908 | 0.7229 | 0.7084 | 0.5904 | 0.7253 | 0.8707 |
| Wheat | 21 | 41 | 143 | 205 | 0.9837 | 0.9767 | 0.9837 | 0.9557 | 0.9767 | 0.9697 | 0.9837 |
| Woods | 127 | 253 | 885 | 1265 | 0.9544 | 0.9529 | 0.9473 | 0.9416 | 0.9529 | 0.9473 | 0.9623 |
| Bldg-Grass-Trees-Drives | 39 | 77 | 270 | 386 | 0.7037 | 0.6457 | 0.6667 | 0.6160 | 0.6321 | 0.5901 | 0.6778 |
| Stone-steel towers | 9 | 19 | 65 | 93 | 0.9590 | 0.9487 | 0.9333 | 0.9077 | 0.9436 | 0.9436 | 0.9897 |
| Total | 1032 | 2051 | 7166 | 10249 | — | — | — | — | — | — | — |
| AA | — | — | — | — | 0.8282 | 0.7981 | 0.8207 | 0.8008 | 0.7954 | 0.8477 | 0.8935 |
| OA | — | — | — | — | 0.8440 | 0.8169 | 0.8310 | 0.8186 | 0.8107 | 0.8521 | 0.8895 |
| Kappa | — | — | — | — | 0.8216 | 0.7906 | 0.8068 | 0.7925 | 0.7834 | 0.8310 | 0.8738 |

4) *KSC:* The data set was taken in March 1996, over Kennedy Space Center in the United States, with 512 × 614 pixels. There are 176 bands and 5211 samples containing 13 classes available. The false color image consisting of bands 50, 30, and 20 and the corresponding ground truth map are, respectively, provided in Fig. 1(d) and (h).

### B. Comparison Methods

To measure the quality of the proposed method, we compare it with several unsupervised and semisupervised BS models, such as ISSC [27], optimal clustering framework (OCF) [55], scalable one-pass self-representation learning (SOPSRL) [56], optimal neighboring reconstruction (ONR) [57], FCGL [29]. A standard SVMs [58] classifier is used during the experiment. In the SVM classification method, the parameters used are set as follows, i.e., 1 for "standardize;" "rbf" for "kernelFunction;" "10" for "BoxConstraint;" "auto" for "KernelScale;" "onevsall" for "Coding". Other parameters are set to default.

1) *Baseline:* the results obtained by SVM with all the bands are regarded as the baseline.
2) *ISSC:* ISSC [27] assumes that each band is drawn from a union of low-dimensional subspaces and each band can be sparsely represented by other bands in its subspace. ISSC algorithm belongs to a sparse representation-based unsupervised model with constructing a similarity matrix with sparse coefficient band vectors.
3) *OCF:* OCF [55] is a clustering-based unsupervised BS method, which can search for the optimal clustering structure on HSI and select the discriminative bands under achieved clustering structure.
4) *SOPSRL:* SOPSRL [56] embeds BS into a scalable self-representation learning, which is formulated as an adaptive linear combination of regression-based loss functions, with the row sparsity constraint. SOPSRL is also an unsupervised BS method.

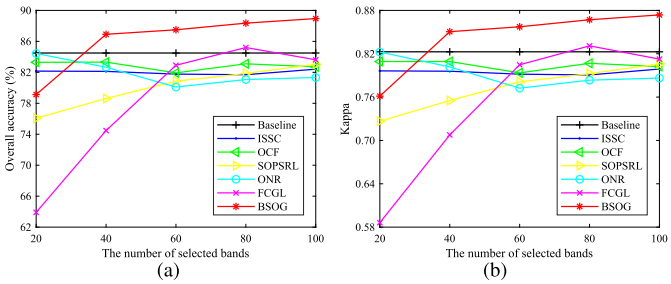



Fig. 2. Relationship between the classification results and the number of selected bands on the Indian Pines data set. (a) OA. (b) Kappa.

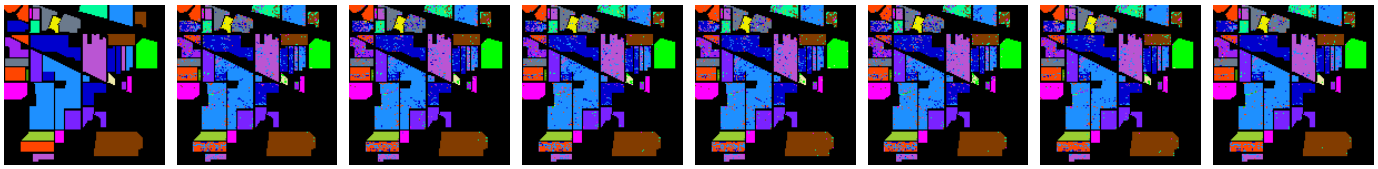
5) *ONR:* ONR [57] exploits the neighboring structure of HSI to select the discriminative band, which belongs to the unsupervised method.
6) *FCGL:* FCGL [29] is a semisupervised BS method, which utilizes labeled and unlabeled samples by combining Fisher's criteria and Graph Laplacian.

### C. Classification Performance With Different Sizes of Band Subset

In this experiment, we compare BSOG with the other five methods in terms of classification accuracy on four data sets with a different number of bands. The selected band numbers are $\{20, 40, \ldots, 100\}$, respectively, on these data sets. The experiments are repeated 10 times to calculate the average results. The parameters of BSOG are specially selected to make the result achieve best, which will be discussed later.

*1) Classification Results on the Indian Pines Data Set:* First, classification experiments are conducted on the Indian Pines data set to compare the performance of different algorithms. 10% and 20% of the samples from each category are randomly chosen as the labeled and unlabeled training sets, while the remaining samples are used in the test part. Note that, for very small classes, we take at least five labeled samples per class (Alfalfa, Grass/ Pasturemowed, and Oats). The detailed

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 11:53:01 UTC from IEEE Xplore. Restrictions apply.



(a) (b) (c) (d) (e) (f) (g) (h)

Fig. 3. Classification maps of Indian image with different approaches. (a) Ground. (b) Baseline. (c) ISSC. (d) OCF. (e) SOPSRL. (f) ONR. (g) FCGL. (h) BSOG.

TABLE II

CLASSIFICATION RESULTS ACHIEVED USING 80 SELECTED BANDS FROM THE SALINAS DATA SET

| Class | Labeled | Unlabeled | Test | Samples | Baseline | ISSC | OCF | SOPSRL | ONR | FCGL | BSOG |
|---|---|---|---|---|---|---|---|---|---|---|---|
| Brocoli_green_weeds_1 | 20 | 40 | 1949 | 2009 | 0.9841 | 0.9810 | 0.9903 | 0.9892 | 0.9885 | 0.9889 | 0.9849 |
| Brocoli_green_weeds_2 | 37 | 75 | 3614 | 3726 | 0.9963 | 0.9961 | 0.9961 | 0.9970 | 0.9966 | 0.9938 | 0.9916 |
| Fallow | 20 | 40 | 1916 | 1976 | 0.9849 | 0.9650 | 0.9788 | 0.9816 | 0.9857 | 0.9833 | 0.9732 |
| Fallow_rough_plow | 14 | 28 | 1352 | 1394 | 0.9914 | 0.9931 | 0.9914 | 0.9943 | 0.9943 | 0.9901 | 0.9909 |
| Fallow_smooth | 27 | 54 | 2597 | 2678 | 0.9772 | 0.9693 | 0.9741 | 0.9775 | 0.9815 | 0.9777 | 0.9791 |
| Stubble | 40 | 80 | 3839 | 3959 | 0.9985 | 0.9990 | 0.9984 | 0.9985 | 0.9986 | 0.9981 | 0.9977 |
| Celery | 36 | 72 | 3471 | 3579 | 0.9955 | 0.9958 | 0.9956 | 0.9955 | 0.9953 | 0.9952 | 0.9947 |
| Grapes_untrained | 113 | 225 | 10933 | 11271 | 0.7975 | 0.7773 | 0.7945 | 0.7947 | 0.8053 | 0.7744 | 0.8144 |
| Soil_vinyard_develop | 62 | 124 | 6017 | 6203 | 0.9963 | 0.9971 | 0.9947 | 0.9966 | 0.9929 | 0.9911 | 0.9919 |
| Corn_senesced_green_weeds | 33 | 65 | 3180 | 3278 | 0.9151 | 0.9079 | 0.9140 | 0.9265 | 0.9202 | 0.9065 | 0.9020 |
| Lettuce_romaine_4wk | 11 | 21 | 1036 | 1068 | 0.9466 | 0.9202 | 0.9427 | 0.9424 | 0.9562 | 0.9466 | 0.9054 |
| Lettuce_romaine_5wk | 19 | 39 | 1869 | 1927 | 0.9963 | 0.9923 | 0.9936 | 0.9963 | 0.9961 | 0.9923 | 0.9980 |
| Lettuce_romaine_6wk | 9 | 18 | 889 | 916 | 0.9835 | 0.9839 | 0.9843 | 0.9816 | 0.9820 | 0.9820 | 0.9790 |
| Lettuce_romaine_7wk | 11 | 21 | 1038 | 1070 | 0.9313 | 0.9226 | 0.9274 | 0.9300 | 0.9351 | 0.9268 | 0.9101 |
| Vinyard_untrained | 73 | 145 | 7050 | 7268 | 0.6488 | 0.6233 | 0.6462 | 0.6751 | 0.6790 | 0.6357 | 0.6704 |
| Vinyard_vertical_trellis | 18 | 36 | 1753 | 1807 | 0.9810 | 0.9816 | 0.9802 | 0.9823 | 0.9785 | 0.9764 | 0.9778 |
| Total | 543 | 1083 | 52503 | 54129 | — | — | — | — | — | — | — |
| AA | — | — | — | — | 0.9447 | 0.9411 | 0.9423 | 0.9445 | 0.9455 | 0.9428 | 0.9482 |
| OA | — | — | — | — | 0.8969 | 0.8924 | 0.8925 | 0.8968 | 0.8970 | 0.8924 | 0.9060 |
| Kappa | — | — | — | — | 0.8852 | 0.8802 | 0.8803 | 0.8851 | 0.8853 | 0.8801 | 0.8953 |

description of selected labeled, unlabeled, and testing samples of each class is shown in Table I.

Fig. 2(a) and (b) show the OA and Kappa versus the number of bands, respectively. From Fig. 2, we have the following observations. First, when the selected band numbers are greater than or equal to 40, our BSOG method performs much better than the other algorithms. Besides, for most BS models (OCF, SOPSRL, ONR, and FCGL), the OA and $\kappa$ do not always increase with the number of bands, which illustrates that more band numbers do not always lead to better classification results.

To obtain an intuitive classification performance evaluation, we fix the number of bands as 80. The corresponding classification maps are given in Fig. 3. It can be seen that our BSOG algorithm produces better classification maps than the other methods. The corresponding detailed classification accuracy of each class, AAs, OAs, and Kappa is described in Table I. From Table I, we can find that our method performs much better than the other approaches on the Indian Pines data set. The OA obtained by BSOG is 88.95%, 4.55% higher than the baseline and 3.74% higher than the FCGL method. The Kappa obtained by BSOG is 0.8738, 0.0522 higher than the baseline and 0.0428 higher than the FCGL method. Then, the corresponding time cost is recorded in Table VI. From this, we can find that our BSOG algorithm spends a little bit more time than other methods because of the constructing graph model and iterative optimization process in our method.

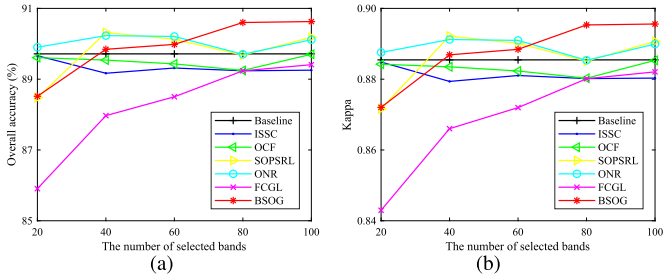



Fig. 4. Relationship between the classification results and the number of selected bands on the Salinas data set. (a) OA. (b) Kappa.

*2) Classification Results on the Salinas Data Set:* Second, the same experiments are operated on the Salinas data set. Since the amount of data in this database is relatively large, with a total of 54 129 samples, we select only 1% of the labeled samples and 2% of the unlabeled samples in the semi-supervised situation for the sake of computational efficiency. The remaining data are regarded as testing samples of SVM classifier in the classification phase. Note that, even 1% of the labeled samples are selected for each category, the number of samples selected is enough. The detailed selection of each class is shown in Table II.

The performance curves are shown in Fig. 4(a) and (b). From Fig. 4, we can find that our BSOG method obtains higher OA and Kappa when the number of selected bands is large than 80.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 11:53:01 UTC from IEEE Xplore. Restrictions apply.

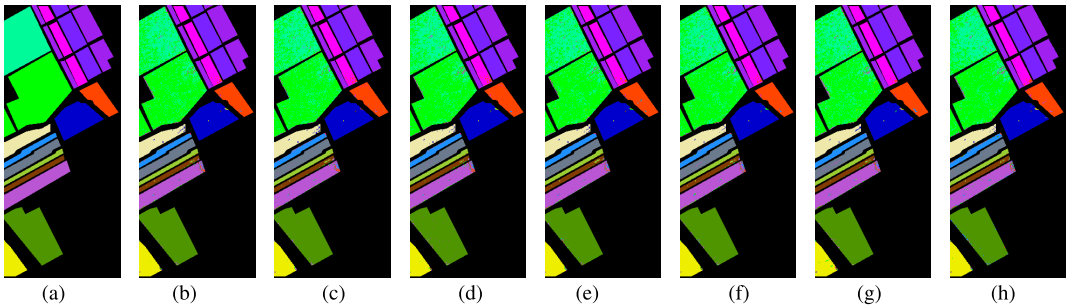



Fig. 5. Classification maps of Salinas image with different approaches. (a) Ground. (b) Baseline. (c) ISSC. (d) OCF. (e) SOPSRL. (f) ONR. (g) FCGL. (h) BSOG.

TABLE III
CLASSIFICATION RESULTS ACHIEVED USING 80 SELECTED BANDS ON BOTSWANA DATA SET

| Class | Labeled | Unlabeled | Test | Samples | Baseline | ISSC | OCF | SOPSRL | ONR | FCGL | BSOG |
|---|---|---|---|---|---|---|---|---|---|---|---|
| Water | 27 | 54 | 189 | 270 | 1 | 1 | 1 | 1 | 1 | 1 | 1 |
| Hippo grass | 10 | 20 | 71 | 101 | 0.9953 | 0.9906 | 0.9906 | 0.9906 | 0.9906 | 1 | 0.9812 |
| Floodplain grasses1 | 25 | 50 | 176 | 251 | 0.9792 | 0.9754 | 0.9792 | 0.9792 | 0.9848 | 0.9848 | 0.9905 |
| Floodplain grasses2 | 22 | 43 | 150 | 215 | 0.9556 | 0.9600 | 0.9533 | 0.9600 | 0.9489 | 0.9422 | 0.9778 |
| Reeds1 | 27 | 54 | 188 | 269 | 0.8741 | 0.8564 | 0.8706 | 0.8617 | 0.8688 | 0.8652 | 0.8493 |
| Riparian | 27 | 54 | 188 | 269 | 0.8156 | 0.8067 | 0.7943 | 0.7677 | 0.7677 | 0.7766 | 0.7855 |
| Firescar2 | 26 | 52 | 181 | 259 | 0.9890 | 0.9853 | 0.9871 | 0.9871 | 0.9890 | 0.9890 | 0.9853 |
| Island interior | 20 | 41 | 142 | 203 | 0.9883 | 0.9789 | 0.9859 | 0.9859 | 0.9812 | 0.9789 | 0.9906 |
| Acacia woodlands | 31 | 63 | 220 | 314 | 0.9227 | 0.9091 | 0.9106 | 0.9045 | 0.9106 | 0.8955 | 0.9273 |
| Acacia shrublands | 25 | 50 | 173 | 248 | 0.9403 | 0.9326 | 0.9403 | 0.9364 | 0.9461 | 0.9287 | 0.9692 |
| Acacia grasslands | 31 | 61 | 213 | 305 | 0.9484 | 0.9327 | 0.9499 | 0.9405 | 0.9452 | 0.9468 | 0.9390 |
| Short mopane | 18 | 36 | 127 | 181 | 0.9423 | 0.9291 | 0.9423 | 0.9449 | 0.9396 | 0.9370 | 0.9291 |
| Mixed mopane | 27 | 54 | 187 | 268 | 0.9180 | 0.9198 | 0.9216 | 0.9180 | 0.9251 | 0.9144 | 0.9180 |
| Exposed soils | 10 | 19 | 66 | 95 | 1 | 1 | 1 | 1 | 1 | 1 | 1 |
| Total | 326 | 651 | 2271 | 3248 | — | — | — | — | — | — | — |
| AA | — | — | — | — | 0.9457 | 0.9437 | 0.9433 | 0.9415 | 0.9422 | 0.9402 | 0.9473 |
| OA | — | — | — | — | 0.9393 | 0.9371 | 0.9366 | 0.9348 | 0.9354 | 0.9331 | 0.9409 |
| Kappa | — | — | — | — | 0.9331 | 0.9319 | 0.9313 | 0.9294 | 0.9301 | 0.9275 | 0.9360 |

When the number of selected band is 80, the corresponding classification maps are given in Fig. 5. From Fig. 5, we can observe that the BSOG method can obtain a smoother classification map. The corresponding detailed classification accuracy of each class, AAs, OAs, and Kappa is shown in Table II. It can be seen that OA obtained by BSOG is 90.60%, 0.91% higher than the baseline and 1.36% higher than the FCGL method. The Kappa obtained by BSOG is 0.8953, 0.0101 higher than the baseline and 0.0152 higher than the FCGL method. Table VI presents the corresponding time costs of different approaches. From this, we can find that our BSOG algorithm is slightly time-consuming compared to other methods, but the difference is marginal with the same quantitative level.

*3) Classification Results on the Botswana Data Set:* Third, the Botswana data set is also used to verify the effectiveness of the BSOG algorithm. The same sample selection is as Indian Pines data set. 10% and 20% of the samples from each category are randomly chosen as the labeled and unlabeled training sets, while the remaining samples are used in the test part. The detailed description of sample selection is listed in Table III.

The relationship between the classification results and the number of bands is shown in Fig. 6. The detailed classification accuracy of each class, AAs, OAs, and Kappa with the SVM classifier after selecting 80 bands by different BS models is given in Table III. From Fig. 6 and Table III, we can find that the results obtained by our BSOG method can exceed the baseline when the number of bands is large than 80, while the others cannot. It also proves that our BSOG algorithm has certain advantages among the BS methods.

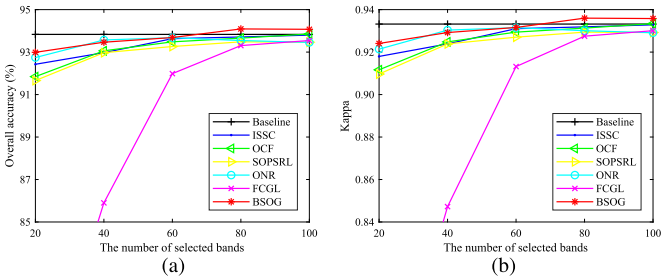



Fig. 6. Relationship between the classification results and the number of selected bands on the Botswana data set. (a) OA. (b) Kappa.

*4) Classification Results on the KSC Data Set:* Finally, the superiority of the proposed BSOG model is tested on KSC data set and the experimental setup is kept the same as that of Botswana data set. 10% and 20% of the samples from each category are randomly chosen as the labeled and unlabeled training sets, while the remaining samples are as the test samples. The corresponding description is shown in Table IV.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 11:53:01 UTC from IEEE Xplore. Restrictions apply.

TABLE IV
CLASSIFICATION RESULTS ACHIEVED USING 80 SELECTED BANDS ON KSC DATA SET

| Class | Labeled | Unlabeled | Test | Samples | Baseline | ISSC | OCF | SOPSRL | ONR | FCGL | BSOG |
|---|---|---|---|---|---|---|---|---|---|---|---|
| Scrub | 76 | 152 | 533 | 761 | 0.9675 | 0.9637 | 0.9606 | 0.9700 | 0.9681 | 0.9650 | 0.9694 |
| Willow swamp | 24 | 49 | 170 | 243 | 0.9118 | 0.8882 | 0.9059 | 0.9275 | 0.9078 | 0.9216 | 0.9137 |
| Cabbage palm hammock | 26 | 51 | 179 | 256 | 0.9311 | 0.9125 | 0.9236 | 0.9162 | 0.9311 | 0.9367 | 0.9032 |
| Cabbage palm/oak | 25 | 50 | 177 | 252 | 0.7458 | 0.7194 | 0.7213 | 0.7307 | 0.7514 | 0.7495 | 0.7891 |
| Slash pine | 16 | 32 | 113 | 161 | 0.7552 | 0.7139 | 0.7227 | 0.7434 | 0.7788 | 0.7375 | 0.7257 |
| Oak/broadleaf hammock | 23 | 46 | 160 | 229 | 0.7333 | 0.7125 | 0.7000 | 0.7229 | 0.7417 | 0.7417 | 0.7604 |
| Hardwood swamp | 11 | 21 | 73 | 105 | 0.8904 | 0.8630 | 0.8721 | 0.8904 | 0.8813 | 0.8858 | 0.8402 |
| Graminoid marsh | 43 | 86 | 302 | 431 | 0.9338 | 0.9349 | 0.9305 | 0.9382 | 0.9459 | 0.9415 | 0.9492 |
| Spartina marsh | 52 | 104 | 364 | 520 | 0.9734 | 0.9707 | 0.9698 | 0.9744 | 0.9789 | 0.9753 | 0.9734 |
| Cattail marsh | 40 | 81 | 283 | 404 | 0.9564 | 0.9529 | 0.9588 | 0.9729 | 0.9494 | 0.9717 | 0.9600 |
| Salt marsh | 42 | 84 | 293 | 419 | 0.9829 | 0.9795 | 0.9829 | 0.9795 | 0.9841 | 0.9875 | 0.9875 |
| Mud flats | 50 | 101 | 352 | 503 | 0.9583 | 0.9593 | 0.9697 | 0.9763 | 0.9631 | 0.9763 | 0.9773 |
| Water | 93 | 185 | 649 | 927 | 0.9969 | 0.9969 | 0.9969 | 1 | 0.9979 | 0.9995 | 0.9964 |
| Total | 521 | 1042 | 3648 | 5211 | — | — | — | — | — | — | — |
| AA | — | — | — | — | 0.9028 | 0.9017 | 0.8949 | 0.8961 | 0.9000 | 0.8976 | 0.9062 |
| OA | — | — | — | — | 0.9370 | 0.9366 | 0.9324 | 0.9358 | 0.9350 | 0.9355 | 0.9402 |
| Kappa | — | — | — | — | 0.9298 | 0.9294 | 0.9247 | 0.9285 | 0.9276 | 0.9281 | 0.9334 |

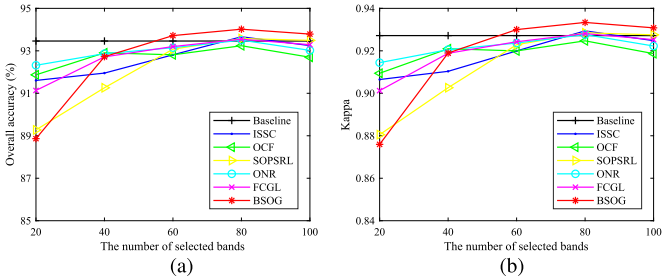



Fig. 7. Relationship between the classification results and the number of selected bands on the KSC data set. (a) OA. (b) Kappa.

Then Fig. 7 presents the relationship between the classification results (OA and Kappa) and the number of selected bands. When the number of bands is fixed as 80, the corresponding results are recorded in Table IV. From them, we can observe that the OA and Kappa obtained by our BSOG method can exceed the baseline when the number of selecting bands is between 60 and 100, while the others cannot.

We list the number of selected bands for all the compared methods in Table V on different data sets when the Kappa achieves best. Besides, for the Indian Pines data set, when the number of selected bands is 40, the performance is good. We list the selected 40 bands in Table V. For the other three data sets, when the number of selected bands fixes 80, the performance is good, thus we list the selected 80 bands in Table V.

### *D. Classification Performance With Different Sizes of Training Samples*

In this experiment, we compare our BSOG algorithm with the other five methods in terms of classification accuracy on four data sets by changing the percentage of labeled samples. For three small data sets (Indian Pines, Botswana, and KSC), the percentage of labeled samples is $\{5, 10, \ldots, 20\}$, while that of unlabeled is fixed as 20. For the large data set (Salinas), the percentage of labeled samples varies $\{0.5, 1, \ldots, 2\}$, while that of unlabeled is fixed as 2. Note that, in exceptional cases,

TABLE V
SELECTED BANDS OF EACH DATA

| Data set | Band |
|---|---|
| Indian Pines | 122, 123, 121, 124, 120, 125, 119, 126, 118, 127, 117, 116, 128, 129, 130, 115, 131, 114, 150, 148, 149, 179, 160, 98, 154, 38, 153, 187, 142, 110, 95, 71, 37, 77, 59, 163, 13, 22, 106, 60 |
| Salinas | 45, 53, 52, 76, 46, 72, 56, 74, 48, 73, 54, 50, 47, 55, 44, 71, 77, 75, 42, 51, 57, 58, 49, 70, 123, 68, 78, 67, 69, 60, 30, 28, 119, 117, 118, 121, 43, 41, 116, 66, 141, 31, 137, 24, 124, 59, 160, 22, 88, 125, 14, 63, 23, 15, 180, 32, 122, 29, 172, 130, 35, 159, 13, 177, 61, 18, 89, 126, 112, 93, 26, 132, 65, 127, 136, 194, 139, 21, 176, 91 |
| Botswana | 88, 91, 75, 100, 102, 74, 60, 69, 70, 72, 78, 92, 71, 29, 134, 130, 116, 137, 143, 141, 133, 123, 135, 120, 131, 124, 129, 127, 121, 144, 126, 132, 125, 1, 145, 139, 136, 12, 142, 14, 37, 140, 21, 11, 19, 4, 40, 2, 20, 119, 33, 36, 16, 17, 31, 138, 15, 5, 128, 10, 8, 6, 28, 118, 38, 9, 39, 7, 23, 18, 122, 13, 24, 34, 45, 112, 113, 35, 26, 32 |
| KSC | 62, 63, 64, 65, 66, 67, 68, 75, 69, 74, 70, 71, 73, 72, 139, 104, 176, 106, 5, 8, 2, 9, 108, 7, 102, 6, 112, 107, 103, 109, 170, 110, 101, 113, 99, 116, 13, 4, 11, 117, 122, 10, 120, 115, 105, 111, 12, 17, 124, 14, 118, 3, 30, 127, 119, 16, 18, 26, 98, 114, 32, 23, 19, 129, 125, 15, 128, 21, 134, 126, 135, 100, 121, 31, 123, 27, 142, 132, 147, 33 |

for very small classes, we take a minimum of five labeled samples per class. Then, the number of selected bands is fixed as 80 on these data sets. The experiments are repeated 10 times to calculate the average results. All the parameters in the six methods are the same as the experiment in Section V-C.

*1) Effect on the Indian Pines Data Set:* First, the classification experiments are conducted on the Indian Pines data set to observe the effect of the labeled samples. The percentage of labeled samples varies from 5 to 20 and that of unlabeled samples is fixed at 20. The corresponding OA and Kappa curves are shown in Fig. 8. From this, we can find that the OA and Kappa obtained by the BSOG model are much higher than the baseline in a different situation, while the other algorithms are lower than that of baseline.

*2) Effect on the Salinas Data Set:* Then, we conduct experiments on the Salinas data set to observe the effect of different percentages of labeled samples. The same reason

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 11:53:01 UTC from IEEE Xplore. Restrictions apply.

TABLE VI

COMPARISON ON TIME COSTS (SECONDS) OF DIFFERENT APPROACHES ON FOUR DATA SETS

| Data set | Baseline | ISSC | OCF | SOPSRL | ONR | FCGL | BSOG |
|---|---|---|---|---|---|---|---|
| Indian Pines | 4.4459 | 3.8975 | 6.2957 | 4.0458 | 4.2465 | 3.8773 | 22.8468 |
| Salinas | 6.6785 | 4.2773 | 6.8023 | 3.7968 | 4.3899 | 3.7433 | 5.7133 |
| Botswana | 0.5830 | 0.6281 | 1.7707 | 0.5478 | 0.8599 | 0.4667 | 1.1165 |
| KSC | 0.8995 | 0.9389 | 2.8085 | 1.1014 | 1.4054 | 0.7650 | 3.0271 |

TABLE VII

BEST KAPPA AND CORRESPONDING BANDS OF DIFFERENT APPROACHES ON FOUR DATA SETS

| Data set | Baseline | ISSC | OCF | SOPSRL | ONR | FCGL | BSOG |
|---|---|---|---|---|---|---|---|
| Indian Pines | 0.8216 | 0.7989(100) | 0.8093(40) | 0.8059(100) | 0.8223(20) | 0.8310(80) | 0.8738(100) |
| Salinas | 0.8852 | 0.8848(20) | 0.8853(100) | 0.8922(40) | 0.8912(40) | 0.8821(100) | 0.8956(100) |
| Botswana | 0.9331 | 0.9328(100) | 0.9334(100) | 0.9294(80) | 0.9315(60) | 0.9301(100) | 0.9360(80) |
| KSC | 0.9298 | 0.9294(80) | 0.9247(80) | 0.9285(80) | 0.9276(80) | 0.9281(80) | 0.9334(80) |

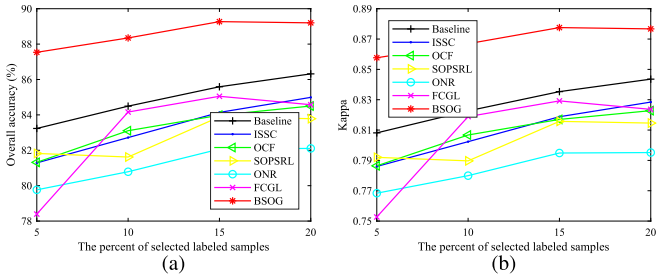



Fig. 8. Relationship between the classification results and the percent of selected labeled samples on the Indian Pines data set. (a) OA. (b) Kappa.

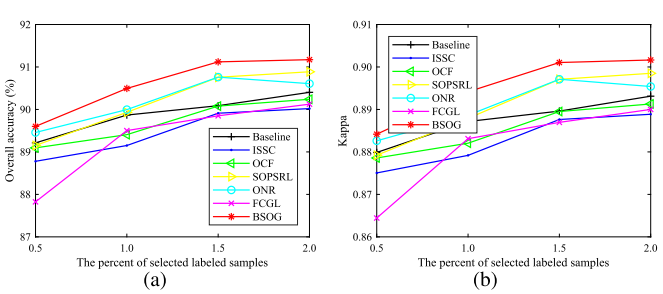



Fig. 9. Relationship between the classification results and the percent of selected labeled samples on the Salinas data set. (a) OA. (b) Kappa.

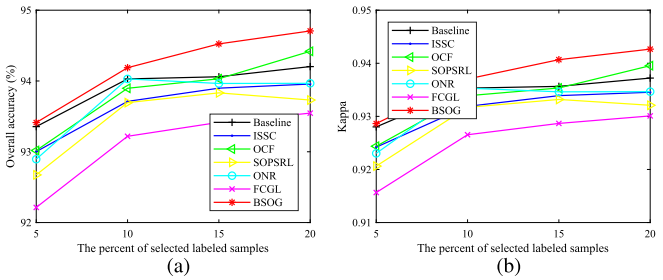



Fig. 10. Relationship between the classification results and the percent of selected labeled samples on the Botswana data set. (a) OA. (b) Kappa.

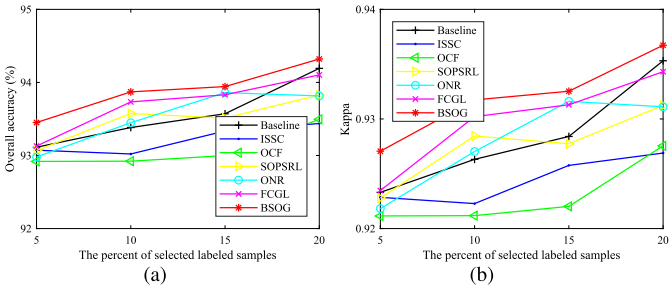



Fig. 11. Relationship between the classification results and the percent of selected labeled samples on the KSC data set. (a) OA. (b) Kappa.

as Section V-C2, we select only $\{0.5\%, 1\%, \ldots, 2\%\}$ labeled samples to improve computational efficiency. The results are shown in Fig. 9. We can find that only SOPSRL, ONR, and BSOG algorithms can exceed the baseline. Among the three BS methods, BSOG performs best in different percentage situations in terms of OA and Kappa.

*3) Effect on the Botswana Data Set:* Third, the same experiments are conducted on the Botswana data set. The percentage of labeled samples is $\{5, 10, \ldots, 20\}$, and that of unlabeled samples is 20. The corresponding classification results are presented in Fig. 10. From Fig. 10, we can see that only the results of our BSOG method are better than the baseline in most condition. Even though the result of OCF can exceed the baseline when the percentage of labeled samples is large than 15, its performance is inferior to that of the BSOG algorithm.

*4) Effect on the KSC Data Set:* Finally, the superiority of the proposed BSOG model is tested on the KSC data set and the experimental setup is kept the same as that of the Botswana data set. The corresponding OA and Kappa curve with the percentage of labeled samples are shown in Fig. 11. From Fig. 11, we can find that our BSOG method is still superior to the other six methods, which verifies the effectiveness of our BSOG algorithm again.

## E. Effect of Parameter

In this section, we test the influence of parameters referred to our model on the four data sets. For three small data sets (Indian Pines, Botswana, and KSC), we select 10% labeled and 20% unlabeled samples in the semisupervised situation, while the remaining data are regarded as testing samples. For the

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 11:53:01 UTC from IEEE Xplore. Restrictions apply.

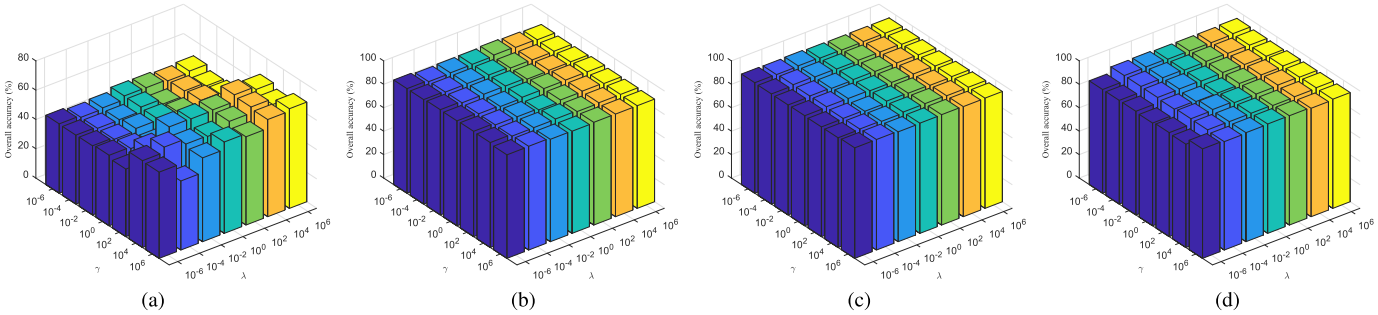



Fig. 12. OA with different $\gamma$ and $\lambda$ values. (a) Indian Pines. (b) Salinas. (c) Botswana. (d) KSC.

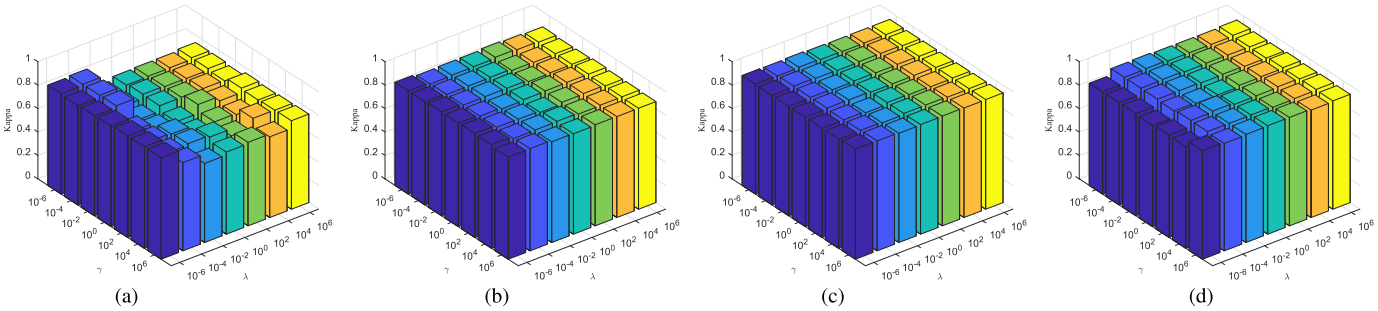



Fig. 13. Kappa with different $\gamma$ and $\lambda$ values. (a) Indian Pines. (b) Salinas. (c) Botswana. (d) KSC.

large data set (Salinas), the percentage of labeled samples is 1, while that of unlabeled is 2. Note that, in exceptional cases, for very small classes, we take a minimum of five labeled samples per class. Then, the number of selected bands are fixed as 80 on these data sets. $\gamma$ and $\lambda$ vary from $10^{-6}$ to $10^{6}$.

For each $\lambda$, $\gamma$ is set to be $\{10^{-6}, 10^{-4}, \ldots, 10^{6}\}$. The corresponding results are shown in Figs. 12 and 13. In these figures, each color represents a certain value of $\lambda$. The height of the histogram represents the value of OA and $\kappa$. The higher the height, the larger the value. From these figures, we can find that both $\gamma$ and $\lambda$ are important to the classification results. To achieve better performance, they should be selected carefully.

## VI. Conclusion

This article proposed a novel semisupervised BS approach, called BSOG, for HSI classification. It performs BS and local structure learning simultaneously by using limited labeled data and abundant unlabeled samples of HSI. Besides, the similarity matrix is constructed in an adaptive way and updated constantly to make the local structure optimal. By analyzing the obtained projection matrix $\boldsymbol{W}$, the most valuable bands can be selected. An efficient optimization algorithm is proposed to solve this model. Comprehensive experiments on 4 well-known HSI data demonstrate the effectiveness of our method.

Also, note that our BSOG algorithm spends a little bit more time than other methods because of the constructing graph model and iterative optimization process in our method. This is also one limitation of our method and can be avoided by adopting the anchor graph method, which is left as our further work.

## Acknowledgment

The authors would like to thank Prof. Sun for providing the code of ISSC from the University of Ningbo University.

## References

[1] A. Ma, Y. Zhong, B. Zhao, H. Jiao, and L. Zhang, "Semisupervised subspace-based DNA encoding and matching classifier for hyperspectral remote sensing imagery," *IEEE Trans. Geosci. Remote Sens.*, vol. 54, no. 8, pp. 4402–4418, Aug. 2016.

[2] G. Camps-Valls, D. Tuia, L. Bruzzone, and J. A. Benediktsson, "Advances in hyperspectral image classification: Earth monitoring with statistical learning methods," *IEEE Signal Process. Mag.*, vol. 31, no. 1, pp. 45–54, Jan. 2014.

[3] J. M. Bioucas-Dias, A. Plaza, G. Camps-Valls, P. Scheunders, N. Nasrabadi, and J. Chanussot, "Hyperspectral remote sensing data analysis and future challenges," *IEEE Geosci. Remote Sens. Mag.*, vol. 1, no. 2, pp. 6–36, Jun. 2013.

[4] R. Wang, F. Nie, and W. Yu, "Fast spectral clustering with anchor graph for large hyperspectral images," *IEEE Geosci. Remote Sens. Lett.*, vol. 14, no. 11, pp. 2003–2007, Nov. 2017.

[5] H. Qv, J. Yin, X. Luo, and X. Jia, "Band dual density discrimination analysis for hyperspectral image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 56, no. 12, pp. 7257–7271, Dec. 2018.

[6] Y. Chen, N. M. Nasrabadi, and T. D. Tran, "Sparse representation for target detection in hyperspectral imagery," *IEEE J. Sel. Topics Signal Process.*, vol. 5, no. 3, pp. 629–640, Jun. 2011.

[7] G. Hughes, "On the mean accuracy of statistical pattern recognizers," *IEEE Trans. Inf. Theory*, vol. 14, no. 1, pp. 55–63, Jan. 1968.

[8] W. Sun and Q. Du, "Graph-regularized fast and robust principal component analysis for hyperspectral band selection," *IEEE Trans. Geosci. Remote Sens.*, vol. 56, no. 6, pp. 3185–3195, Jun. 2018.

[9] X. Zhang, Y. He, N. Zhou, and Y. Zheng, "Semisupervised dimensionality reduction of hyperspectral images via local scaling cut criterion," *IEEE Geosci. Remote Sens. Lett.*, vol. 10, no. 6, pp. 1547–1551, Nov. 2013.

[10] W. Zhang, X. Li, and L. Zhao, "A fast hyperspectral feature selection method based on band correlation analysis," *IEEE Geosci. Remote Sens. Lett.*, vol. 15, no. 11, pp. 1750–1754, Nov. 2018.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 11:53:01 UTC from IEEE Xplore. Restrictions apply.

[11] H. Zhai, H. Zhang, L. Zhang, and P. Li, "Laplacian-regularized low-rank subspace clustering for hyperspectral image band selection," *IEEE Trans. Geosci. Remote Sens.*, vol. 57, no. 3, pp. 1723–1740, Mar. 2019.

[12] J. Feng, L. Jiao, F. Liu, T. Sun, and X. Zhang, "Mutual-information-based semi-supervised hyperspectral band selection with high discrimination, high information, and low redundancy," *IEEE Trans. Geosci. Remote Sens.*, vol. 53, no. 5, pp. 2956–2969, May 2015.

[13] S. Wold, K. Esbensen, and P. Geladi, "Principal component analysis," *Chemometrics Intell. Lab. Syst.*, vol. 2, nos. 1–3, pp. 37–52, 1987.

[14] D. D. Lee and H. S. Seung, "Learning the parts of objects by non-negative matrix factorization," *Nature*, vol. 401, no. 6755, p. 788, 1999.

[15] A. Hyvärinen, J. Karhunen, and E. Oja, *Independent Component Analysis*, vol. 46. Hoboken, NJ, USA: Wiley, 2004.

[16] W. Li, S. Prasad, J. E. Fowler, and L. M. Bruce, "Locality-preserving dimensionality reduction and classification for hyperspectral image analysis," *IEEE Trans. Geosci. Remote Sens.*, vol. 50, no. 4, pp. 1185–1198, Apr. 2012.

[17] S. T. Roweis, "Nonlinear dimensionality reduction by locally linear embedding," *Science*, vol. 290, no. 5500, pp. 2323–2326, Dec. 2000.

[18] B. Guo, S. R. Gunn, R. I. Damper, and J. D. B. Nelson, "Band selection for hyperspectral image classification using mutual information," *IEEE Geosci. Remote Sens. Lett.*, vol. 3, no. 4, pp. 522–526, Oct. 2006.

[19] H. Yang, Q. Du, H. Su, and Y. Sheng, "An efficient method for supervised hyperspectral band selection," *IEEE Geosci. Remote Sens. Lett.*, vol. 8, no. 1, pp. 138–142, Jan. 2011.

[20] J. Feng, L. Jiao, T. Sun, H. Liu, and X. Zhang, "Multiple kernel learning based on discriminative kernel clustering for hyperspectral band selection," *IEEE Trans. Geosci. Remote Sens.*, vol. 54, no. 11, pp. 6516–6530, Nov. 2016.

[21] S. Feng, Y. Itoh, M. Parente, and M. F. Duarte, "Hyperspectral band selection from statistical wavelet models," *IEEE Trans. Geosci. Remote Sens.*, vol. 55, no. 4, pp. 2111–2123, Apr. 2017.

[22] F. Nie, D. Xu, I. W.-H. Tsang, and C. Zhang, "Flexible manifold embedding: A framework for semi-supervised and unsupervised dimension reduction," *IEEE Trans. Image Process.*, vol. 19, no. 7, pp. 1921–1932, Jul. 2010.

[23] F. Nie, S. Xiang, Y. Liu, and C. Zhang, "A general graph-based semi-supervised learning with novel class discovery," *Neural Comput. Appl.*, vol. 19, no. 4, pp. 549–555, Jun. 2010.

[24] J. Li, J. M. Bioucas-Dias, and A. Plaza, "Semisupervised hyperspectral image classification using soft sparse multinomial logistic regression," *IEEE Geosci. Remote Sens. Lett.*, vol. 10, no. 2, pp. 318–322, Mar. 2013.

[25] C.-I. Chang, Q. Du, T.-L. Sun, and M. L. G. Althouse, "A joint band prioritization and band-decorrelation approach to band selection for hyperspectral image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 37, no. 6, pp. 2631–2641, Nov. 1999.

[26] C.-I. Chang and S. Wang, "Constrained band selection for hyperspectral imagery," *IEEE Trans. Geosci. Remote Sens.*, vol. 44, no. 6, pp. 1575–1585, Jun. 2006.

[27] W. Sun, L. Zhang, B. Du, W. Li, and Y. Mark Lai, "Band selection using improved sparse subspace clustering for hyperspectral imagery classification," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 8, no. 6, pp. 2784–2797, Jun. 2015.

[28] Y. Yuan, J. Lin, and Q. Wang, "Dual-clustering-based hyperspectral band selection by contextual analysis," *IEEE Trans. Geosci. Remote Sens.*, vol. 54, no. 3, pp. 1431–1445, Mar. 2016.

[29] L. Chen, R. Huang, and W. Huang, "Graph-based semi-supervised weighted band selection for classification of hyperspectral data," in *Proc. Int. Conf. Audio, Lang. Image Process.*, Nov. 2010, pp. 1123–1126.

[30] H. Su, H. Yang, Q. Du, and Y. Sheng, "Semisupervised band clustering for dimensionality reduction of hyperspectral imagery," *IEEE Geosci. Remote Sens. Lett.*, vol. 8, no. 6, pp. 1135–1139, Nov. 2011.

[31] J. Feng, L. C. Jiao, X. Zhang, and T. Sun, "Hyperspectral band selection based on trivariate mutual information and clonal selection," *IEEE Trans. Geosci. Remote Sens.*, vol. 52, no. 7, pp. 4092–4105, Jul. 2014.

[32] X. Bai, Z. Guo, Y. Wang, Z. Zhang, and J. Zhou, "Semisupervised hyperspectral band selection via spectral–spatial hypergraph model," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 8, no. 6, pp. 2774–2783, Jun. 2015.

[33] X. Cao, C. Wei, Y. Ge, J. Feng, J. Zhao, and L. Jiao, "Semi-supervised hyperspectral band selection based on dynamic classifier selection," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 12, no. 4, pp. 1289–1298, Apr. 2019.

[34] W. Sun and Q. Du, "Hyperspectral band selection: A review," *IEEE Geosci. Remote Sens. Mag.*, vol. 7, no. 2, pp. 118–139, Jun. 2019.

[35] S. Jia, G. Tang, J. Zhu, and Q. Li, "A novel ranking-based clustering approach for hyperspectral band selection," *IEEE Trans. Geosci. Remote Sens.*, vol. 54, no. 1, pp. 88–102, Jan. 2016.

[36] Q. Du and H. Yang, "Similarity-based unsupervised band selection for hyperspectral image analysis," *IEEE Geosci. Remote Sens. Lett.*, vol. 5, no. 4, pp. 564–568, Oct. 2008.

[37] C. Yang, S. Liu, L. Bruzzone, R. Guan, and P. Du, "A feature-metric-based affinity propagation technique for feature selection in hyperspectral image classification," *IEEE Geosci. Remote Sens. Lett.*, vol. 10, no. 5, pp. 1152–1156, Sep. 2013.

[38] Y. Yuan, X. Zheng, and X. Lu, "Discovering diverse subset for unsupervised hyperspectral band selection," *IEEE Trans. Image Process.*, vol. 26, no. 1, pp. 51–64, Jan. 2017.

[39] W. Sun, W. Li, J. Li, and Y. M. Lai, "Band selection using sparse nonnegative matrix factorization with the thresholded Earth's mover distance for hyperspectral imagery classification," *Earth Sci. Informat.*, vol. 8, no. 4, pp. 907–918, Dec. 2015.

[40] B. B. Damodaran, N. Courty, and S. Lefevre, "Sparse Hilbert Schmidt independence criterion and surrogate-kernel-based feature selection for hyperspectral image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 55, no. 4, pp. 2385–2398, Apr. 2017.

[41] R. Archibald and G. Fann, "Feature selection and classification of hyperspectral images with support vector machines," *IEEE Geosci. Remote Sens. Lett.*, vol. 4, no. 4, pp. 674–677, Oct. 2007.

[42] P. Zhong, P. Zhang, and R. Wang, "Dynamic learning of SMLR for feature selection and classification of hyperspectral data," *IEEE Trans. Geosci. Remote Sens.*, vol. 5, no. 2, pp. 280–284, Apr. 2008.

[43] A. Zare and P. Gader, "Hyperspectral band selection and endmember detection using sparsity promoting priors," *IEEE Geosci. Remote Sens. Lett.*, vol. 5, no. 2, pp. 256–260, Apr. 2008.

[44] Y. Zhan, D. Hu, H. Xing, and X. Yu, "Hyperspectral band selection based on deep convolutional neural network and distance density," *IEEE Geosci. Remote Sens. Lett.*, vol. 14, no. 12, pp. 2365–2369, Dec. 2017.

[45] Q. Wang, J. Lin, and Y. Yuan, "Salient band selection for hyperspectral image classification via manifold ranking," *IEEE Trans. Neural Netw. Learn. Syst.*, vol. 27, no. 6, pp. 1279–1289, Jun. 2016.

[46] S. Li, H. Wu, D. Wan, and J. Zhu, "An effective feature selection method for hyperspectral image classification based on genetic algorithm and support vector machine," *Knowl.-Based Syst.*, vol. 24, no. 1, pp. 40–48, Feb. 2011.

[47] J. Zhao, K. Lu, and X. He, "Locality sensitive semi-supervised feature selection," *Neurocomputing*, vol. 71, nos. 10–12, pp. 1842–1849, Jun. 2008.

[48] F. Nie, X. Wang, M. I. Jordan, and H. Huang, "The constrained Laplacian rank algorithm for graph-based clustering," in *Proc. 30th AAAI*, Feb. 2016, pp. 1969–1976.

[49] F. Nie, X. Wang, and H. Huang, "Clustering and projected clustering with adaptive neighbors," in *Proc. 20th ACM SIGKDD Int. Conf. Knowl. Discovery Data Mining (KDD)*, 2014, pp. 977–986.

[50] F. Nie, W. Zhu, and X. Li, "Unsupervised feature selection with structured graph optimization," in *Proc. 30th AAAI*, 2016, pp. 1302–1308.

[51] F. Nie, H. Huang, X. Cai, and C. H. Ding, "Efficient and robust feature selection via joint $\ell_{2,1}$-norms minimization," in *Proc. 23rd NIPS*, 2010, pp. 1813–1821.

[52] T. Pang, F. Nie, J. Han, and X. Li, "Efficient feature selection via $\ell_{2,0}$-norm constrained sparse regression," *IEEE Trans. Knowl. Data Eng.*, vol. 31, no. 5, pp. 880–893, May 2019.

[53] Y. Yang, H. T. Shen, Z. Ma, Z. Huang, and X. Zhou, "$\ell_{2,1}$-norm regularized discriminative feature selection for unsupervised," in *Proc. 22nd IJCAI*, 2011, pp. 1589–1594.

[54] D. Wang, F. Nie, and H. Huang, "Unsupervised feature selection via unified trace ratio formulation and $k$-means clustering (track)," in *Proc. ECML/PKDD*, 2014, pp. 306–321.

[55] Q. Wang, F. Zhang, and X. Li, "Optimal clustering framework for hyperspectral band selection," *IEEE Trans. Geosci. Remote Sens.*, vol. 56, no. 10, pp. 5910–5922, Oct. 2018.

[56] X. Wei, W. Zhu, B. Liao, and L. Cai, "Scalable one-pass self-representation learning for hyperspectral band selection," *IEEE Trans. Geosci. Remote Sens.*, vol. 57, no. 7, pp. 4360–4374, Jul. 2019.

[57] F. Zhang, Q. Wang, and X. Li, "Optimal neighboring reconstruction for hyperspectral band selection," in *Proc. IEEE Int. Geosci. Remote Sens. Symp. (IGARSS)*, Jul. 2018, pp. 4709–4712.

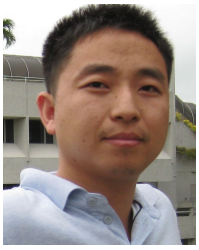
[58] M. A. Hearst, S. T. Dumais, E. Osuna, J. Platt, and B. Scholkopf, "Support vector machines," *IEEE Intell. Syst. Appl.*, vol. 13, no. 4, pp. 18–28, Jul. 1998.


Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 11:53:01 UTC from IEEE Xplore. Restrictions apply.



**Fang He** received the B.S. and M.S. degrees from the High-Tech Institute of Xi'an, Xi'an, China, in 2014 and 2016, respectively, where she is pursuing the Ph.D. degree. She is also pursuing the Ph.D. degree with the Center for OPTical IMagery Analysis and Learning (OPTIMAL), Northwestern Polytechnical University, Xi'an.

Her research interest includes machine learning and its applications.



**Feiping Nie** (Member, IEEE) received the Ph.D. degree in computer science from Tsinghua University, Beijing, China, in 2009.

He is a Full Professor with Northwestern Polytechnical University, Xi'an, China. He has authored or coauthored more than 100 articles in the following journals and conferences: *International Journal of Computer Vision*, the IEEE TRANSACTIONS ON PATTERN ANALYSIS AND MACHINE INTELLIGENCE, the IEEE TRANSACTIONS ON IMAGE PROCESSING, the IEEE TRANSACTIONS ON NEURAL NETWORKS AND LEARNING SYSTEMS, and the IEEE TRANSACTIONS ON KNOWLEDGE AND DATA ENGINEERING. His research interests include machine learning and its applications, such as pattern recognition, data mining, computer vision, image processing, and information retrieval. His articles have been cited more than 10 000 times and the H-index is 57.

Dr. Nie is an Associate Editor or a Program Committee Member for several prestigious journals and conferences.



**Rong Wang** (Member, IEEE) received the B.E. degree in information engineering, the M.E. degree in signal and information processing, and the Ph.D. degree in computer science from the Xi'an Research Institute of Hi-Tech, Xi'an, China, in 2004, 2007, and 2013, respectively, and the Ph.D. degree with the Department of Automation, Tsinghua University, Beijing, China, in 2013.

He is an Associate Professor with the Center for OPTical IMagery Analysis and Learning (OPTIMAL), Northwestern Polytechnical University, Xi'an. His research interests include machine learning and signal processing, together with their applications including pattern recognition, image processing, and computer vision.



**Weimin Jia** received the M.S. and Ph.D. degrees from the Xi'an Research Institute of Hi-Tech, Xi'an, China, in 1998 and 2007, respectively, all in electronic engineering.

She is a Professor with the Department of Communication Engineering, Xi'an Research Institute of Hi-Tech. Her research interests are in array signal processing.



**Fenggan Zhang** received the M.S. and Ph.D. degrees from the Xi'an Research Institute of Hi-Tech, Xi'an, China, in 2009 and 2014, respectively, all in electronic engineering.

He is a Lecturer with the Department of Communication Engineering, Xi'an Research Institute of Hi-Tech. His research interests are the land mobile satellite communication and array antenna.

**Xuelong Li** (Fellow, IEEE) is a Full Professor with the School of Computer Science, Center for OPTical IMagery Analysis and Learning (OPTIMAL), Northwestern Polytechnical University, Xi'an, China.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 11:53:01 UTC from IEEE Xplore. Restrictions apply.